19.5 INCH TIRE VALVE STEM INSPECTION AND SERVICE

AFFECTED VEHICLES: CERTAIN 1999-2001 F-450 AND F-550 SUPER DUTY TRUCKS EQUIPPED WITH 19.5 INCH COMMERCIAL TRUCK TIRES WITH STEEL CORD SIDEWALLS BUILT AT KENTUCKY TRUCK PLANT

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WARNING!

AS WITH ANY AUTOMOTIVE REPAIR SERVICE, PERSONAL PROTECTIVE EQUIPMENT (PPE) SUCH AS EYE PROTECTION MUST BE WORN DURING ALL PHASES OF THIS REPAIR.

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OVERVIEW

This service procedure involves inspecting wheel valve stems for leakage. If a leaking valve stem is found, you will have the option of replacing the tire and valve stem on the affected wheel, or you may sublet the valve stem and tire mount and balance procedure to a truck tire facility. Finally, you will be adjusting the tire inflation pressure to specification on all wheels using the proper safety procedure.

WARNING!

PERMANENT TIRE DAMAGE DUE TO UNDERINFLATION AND/OR OVERLOADING CANNOT ALWAYS BE DETECTED. ANY TIRE KNOWN OR SUSPECTED TO HAVE BEEN RUN AT 80% OR LESS OF NORMAL OPERATING INFLATION PRESSURE AND/OR OVERLOADED, COULD POSSIBLY HAVE PERMANENT STRUCTURAL DAMAGE (STEEL CORD FATIGUE). PLY CORDS WEAKENED BY UNDERINFLATION AND/OR OVERLOADING MAY BREAK ONE AFTER ANOTHER, UNTIL A RUPTURE OCCURS IN THE UPPER SIDEWALL WITH ACCOMPANYING INSTANTANEOUS AIR LOSS AND EXPLOSIVE FORCE. THIS CAN RESULT IN SERIOUS INJURY OR DEATH.

IMPORTANT SAFETY INFORMATION

SERVICING STEEL CORD SIDEWALL TIRES

The 19.5 inch steel cord sidewall tires equipped on F-450 and F-550 Super Duty vehicles are of an ***all steel*** radial construction, and operate at inflation pressures of up to 95 psi. As such, they are NOT light truck tires and must be serviced with the same safety considerations that are used with other medium/heavy truck steel radial tires. Personnel trained, supervised, and equipped in compliance with OSHA Standard 29 CFR - 1910.177 (refer to Attachment IV), must perform tire service, including adjusting tire pressure. For example, during any procedure involving tire inflation, a technician or individual must utilize a remote inflation device and ensure that all persons are clear of the trajectory area. See Figures 1 and 2.

SERVICE RESPONSIBILITIES

If your dealership does not have the equipment to safely replace tires of this size and type, you will be instructed to sublet only the tire replacement and balance portion of this repair to a local truck tire facility. Instructions have been clearly outlined defining which repair steps you, the dealer, will be expected to perform and which you may sublet. Also, at 160 kilometers (100 miles) and 800 kilometers (500 miles) after any wheel has been removed and replaced, the torque on the lug nuts will need to be rechecked. This service is expected to be done by the dealer.

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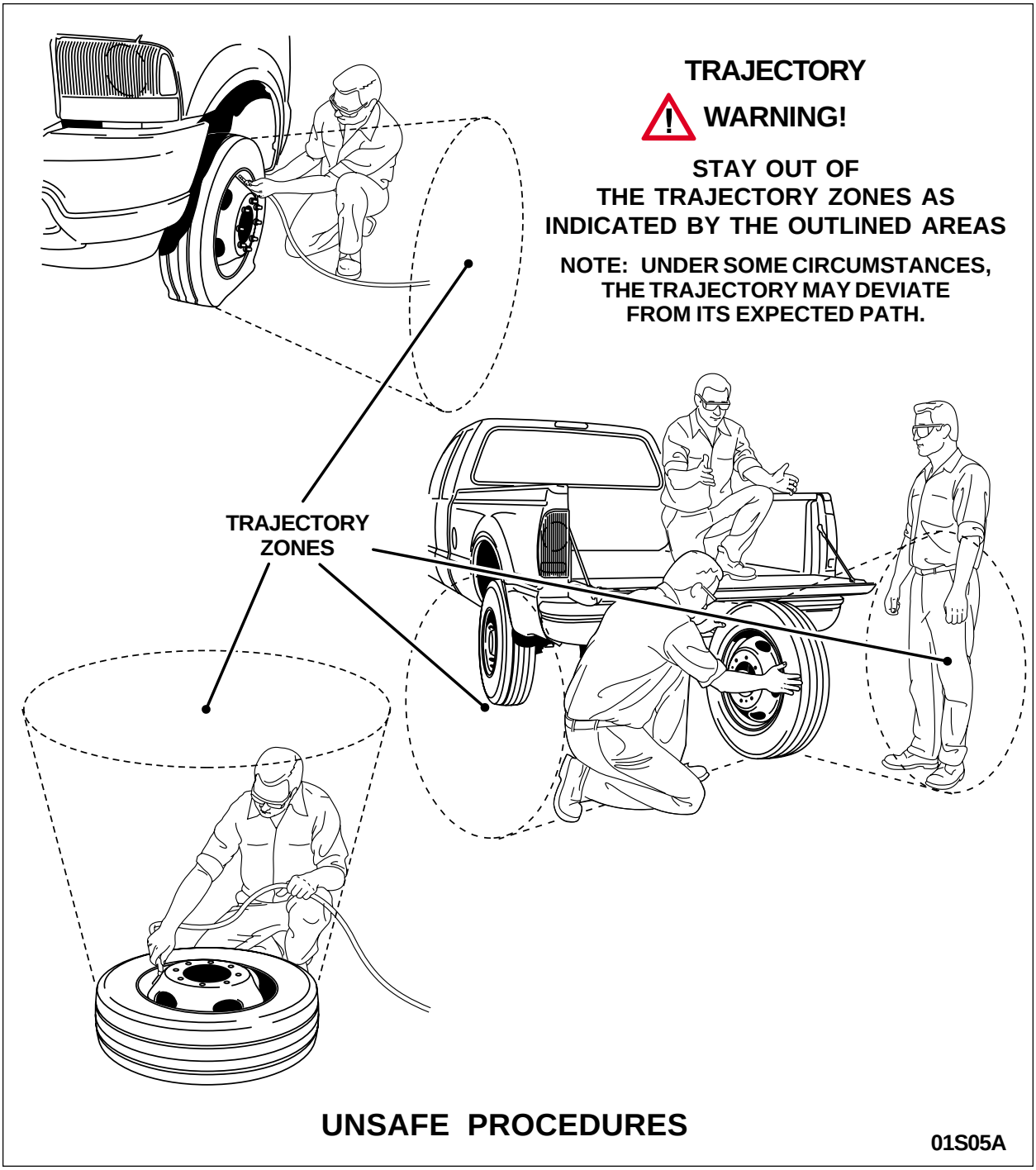
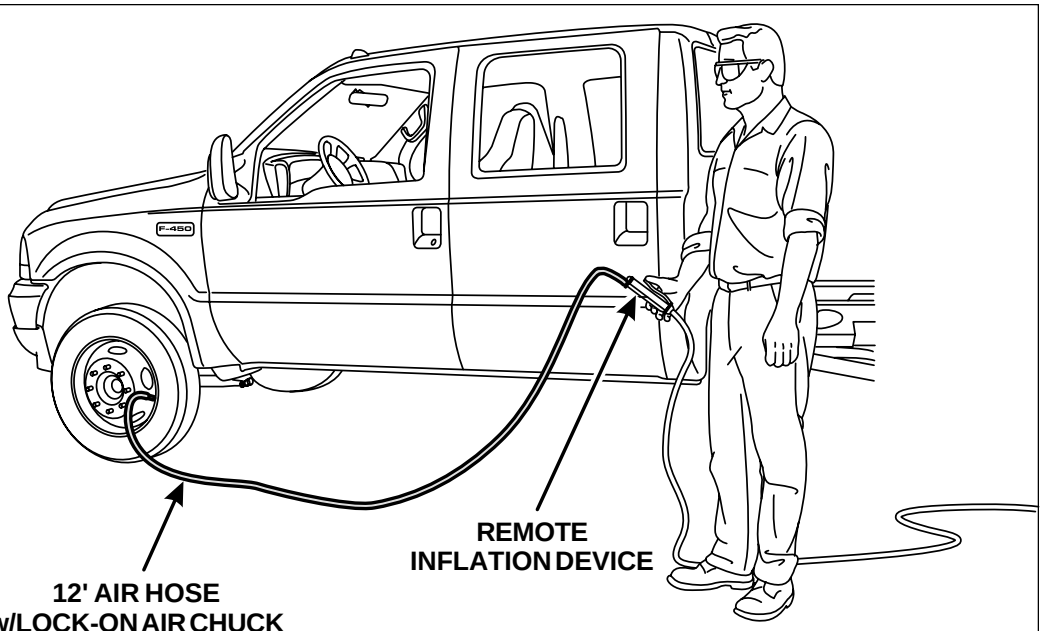


FIGURE 1
UNSAFE TIRE INFLATION AND HANDLING TECHNIQUES

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STAND OUTSIDE TRAJECTORY ZONE WHILE INFLATING TIRE

SAFE PROCEDURE

01S05B

FIGURE 2
SAFE INFLATION TECHNIQUE

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INSPECTION *(Must be completed by Ford or Lincoln Mercury Dealership)*

WARNING!

AS WITH ANY AUTOMOTIVE REPAIR SERVICE, PERSONAL PROTECTIVE EQUIPMENT (PPE) SUCH AS EYE PROTECTION MUST BE WORN DURING ALL PHASES OF THIS REPAIR.

1. Perform inspection of six (6) road tire valve stems, and one (1) spare tire valve stem (if equipped).
   A. Spray the area surrounding the valve stem with a soapy water solution.
   B. **CAUTION: Do not bend the valve stem more than 30° or stem damage may occur.**
   Slowly move the valve in a small circular pattern, stopping rotation if bubbles appear. See Figure 3.

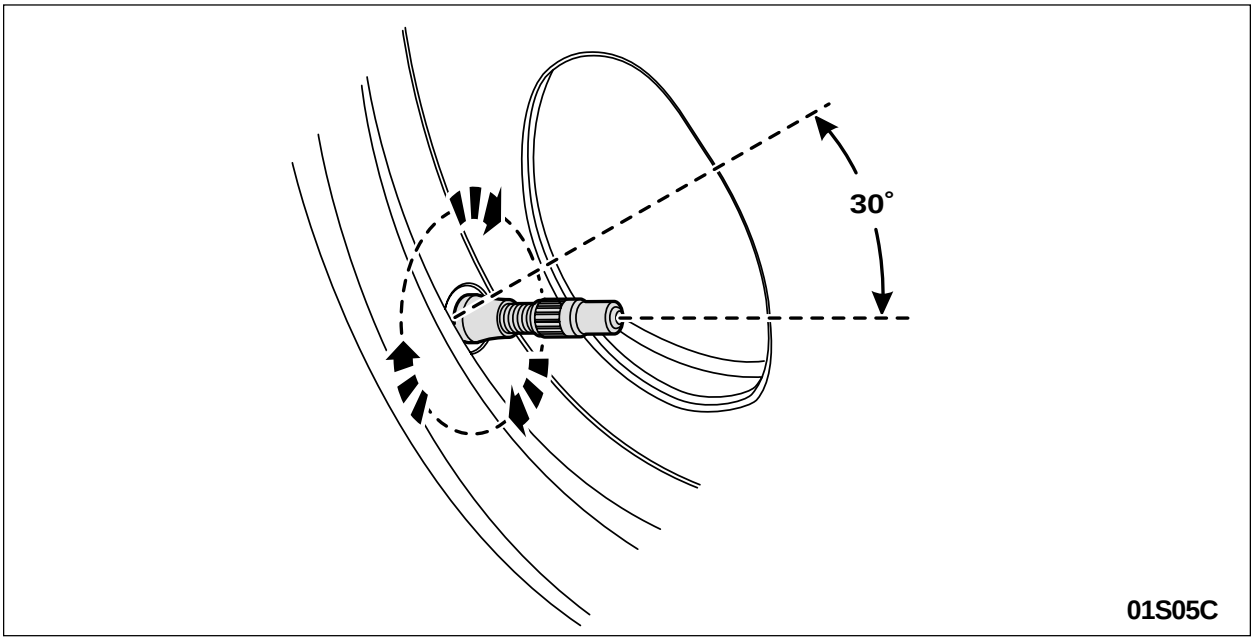


FIGURE 3

   C. A continuous stream of bubbles as the valve is held in this position indicates valve stem leakage.

NOTE: Due to air entry into the area surrounding the valve stem during wheel assembly, a small non-continuous flow of bubbles may be present as you initially move the valve. This is a normal occurrence on some wheels, and does not necessarily denote a valve stem leak. A true valve stem leak will have a repeatable continuous stream of bubbles when the valve stem is held in a particular position.

   D. Repeat this test for all wheels including the spare tire (if equipped).
   E. A continuous flow of bubbles as described above will require that the valve stem and tire be replaced on the affected wheel.

2. If a leaking valve stem is found, proceed to TIRE AND VALVE STEM REPLACEMENT.

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3. On all tires not being replaced, set the tire pressures to specifications as follows:

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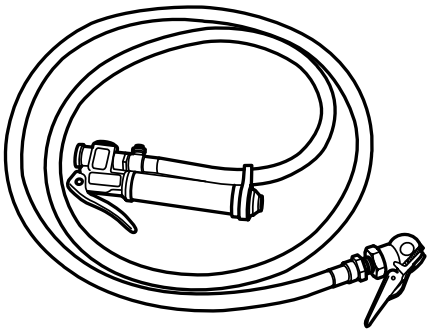
A) Prepare to inflate the tire using:

I. Lock-on air chuck attached to a minimum of 12 feet of hose sufficient to clear the trajectory zone.

II. Remote inflation gauge and air flow control or equivalent combination air pressure gauge and air flow control assembly. See Figure 4.

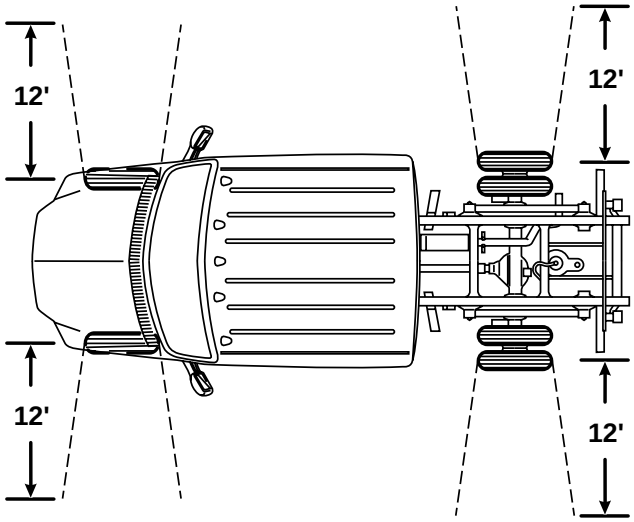
B) Verify that all individuals are outside the trajectory area (12 feet). See Figure 5.

C) Inflate the tire to specification (refer to Safety Certification Label located in the driver's door opening).



REMOTE INFLATION DEVICE W/LOCK-ON AIR CHUCK **01S05D**

FIGURE 4



12' TRAJECTORY ZONES **01S05E**

FIGURE 5

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TIRE AND VALVE STEM REPLACEMENT

- If your dealership is properly equipped to safely replace tires of this size and type, refer to the procedures included in DEALERSHIP TIRE SERVICE — COMPLETE.
- If your dealership does not have the equipment to safely replace tires of this size and type, you are instructed to sublet the tire replacement and balance portion of this repair to a local truck tire facility. Refer to the procedure titled DEALERSHIP/SUBLET TIRE SERVICE. There you will find separate instructions for the dealer (wheel/tire R&I and tire inflation) and for the sublet tire facility (tire replacement, valve stem replacement and tire balance only). **Be sure to also supply a copy of the instructions to the sublet tire facility.**

DEALERSHIP TIRE SERVICE — COMPLETE

ROAD WHEEL TIRE AND VALVE STEM REPLACEMENT

NOTE: Perform this procedure for each wheel with a leaking valve stem.

1. Remove the valve stem core and deflate the affected tire. Then remove the tire from the vehicle.
2. Remove the tire from the wheel, following tire changing equipment manufacturer recommendations.
3. Inspect the wheel for damage. Bent, cracked or otherwise damaged wheels must not be reused or repaired.
4. Replace the valve stem with either a Dill® E-600R or a Schraeder® TR-600HP valve stem.
5. **CAUTION: Do not use a petroleum-based lubricant on the replacement tire. Damage to the bead area of the tire may occur.**

   Lubricate the bead of the ***new*** replacement tire using a commercially available tire lubricant. Then mount the tire on the wheel, working from the valve side.
6. **WARNING: SEAT THE BEAD ONLY AT THIS TIME. DO NOT INFLATE THE TIRE TO SPECIFICATION UNTIL IT IS ON THE VEHICLE AND THE LUG NUTS ARE TIGHTENED TO SPECIFICATION.**

   Inflate tire with enough air pressure to seat the bead (20 psi maximum).
7. Install the valve stem core.
8. Balance wheel and tire assembly.

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9. Reinstall the wheel/tire assembly(s) onto the vehicle. Tighten the lug nuts to 217 Nm (160 lb-ft). See Figure 6.

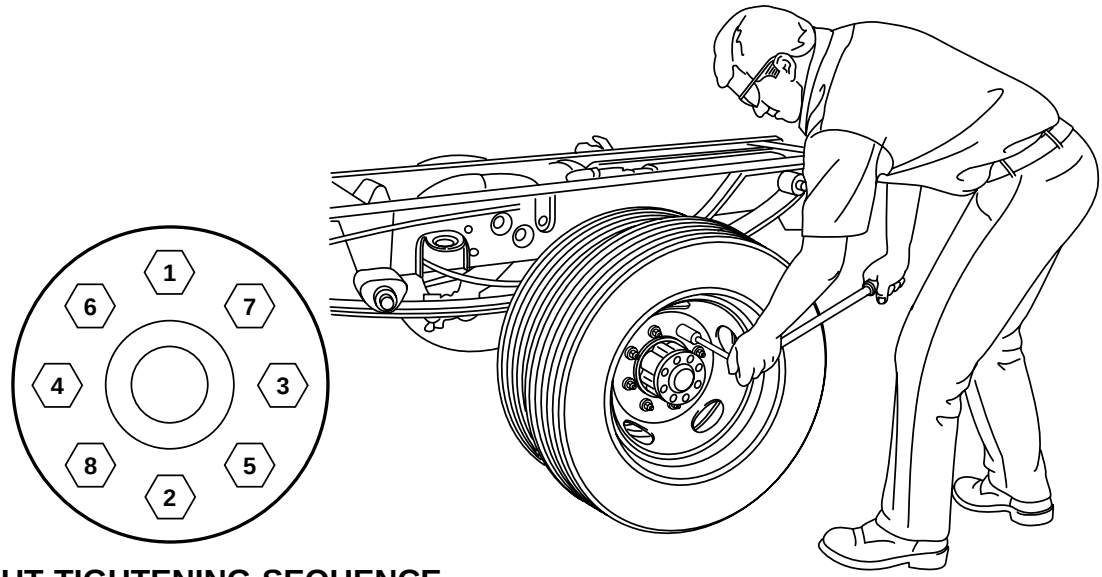


FIGURE 6

10. **WARNING: BE SURE TO READ AND FOLLOW THESE INSTRUCTIONS CAREFULLY. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN SEVERE PERSONAL INJURY OR DEATH.**
    A) Prepare to inflate the tire using:
       I. Lock-on air chuck attached to a minimum of 12 feet of hose sufficient to clear the trajectory zone.
       II. Remote inflation gauge and air flow control or equivalent combination air pressure gauge and air flow control assembly. See Figure 4.
    B) Verify that all individuals are outside the trajectory area (12 feet). See Figure 5.
    C) Verify the wheel/tire assembly is bolted to the axle and the lug nuts are tightened to specification.
    D) Inflate the tire to specification (refer to Safety Certification Label located in the driver's door opening).
11. Following Step 10, adjust the inflation pressure on the remaining wheels that do not have a valve stem leak.
12. Render the old tire unusable by drilling 4 one-inch (approximately 25 mm) holes in the old tire sidewall and cutting the DOT code out with a utility knife.
13. **CAUTION: The torque on the lug nuts must be checked 160 kilometers (100 miles) AND 800 kilometers (500 miles) after tire replacement.**

    Using a suitable torque wrench, check and retighten all lug nuts to 217 Nm (160 lb-ft). See Figure 6.

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SPARE TIRE AND VALVE STEM REPLACEMENT

1. Remove the valve stem core and deflate the tire. Then remove the tire from the spare tire carrier.
2. Remove the tire from the wheel, following tire changing equipment manufacturer recommendations.
3. Inspect the wheel for damage. Bent, cracked or otherwise damaged wheels must not be reused or repaired.
4. Replace the valve stem with either a Dill® E-600R or a Schraeder® TR-600HP valve stem.
5. **CAUTION: Do not use a petroleum-based lubricant on the replacement tire. Damage to the bead area of the tire may occur.**

   Lubricate the bead of the ***new*** replacement tire, using a commercially available tire lubricant. Then mount the tire on the wheel, working from the valve side.
6. **WARNING: DO NOT INFLATE THE TIRE UNLESS IT IS PLACED IN AN APPROVED SAFETY CAGE OR MOUNTED TO THE VEHICLE IN PLACE OF A ROAD WHEEL AND THE LUG NUTS ARE TIGHTENED TO SPECIFICATION. SEAT THE BEAD ONLY AT THIS TIME.**

   Inflate tire with enough air to seat the tire bead (20 psi maximum).
7. Install the valve stem core.
8. Balance the wheel and tire assembly.
9. **WARNING: BE SURE TO READ AND FOLLOW THESE INSTRUCTIONS CAREFULLY. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN SEVERE PERSONAL INJURY OR DEATH.**
   A) Place the tire and wheel assembly into an approved safety cage or mount it to the vehicle in place of a road wheel and tighten the lug nuts to specification.
   B) Prepare to inflate the tire using:
      I. Lock-on air chuck attached to a minimum of 12 feet of hose sufficient to clear the trajectory zone.
      II. Remote inflation gauge and air flow control or equivalent combination air pressure gauge and air flow control assembly. See Figure 4.
   C) Verify that all individuals are outside the trajectory area (12 feet). See Figure 5.
   D) Verify that the wheel/tire assembly is in an approved safety cage or bolted to the axle, and the lug nuts are tightened to specification.
   E) Inflate the tire to specification (refer to Safety Certification Label located in the driver's door opening).
10. Reinstall the spare tire into the spare tire carrier, if necessary.
11. Following Step 9, adjust the inflation pressure on the remaining wheels that do not have a valve stem leak.
12. Render the old tire unusable by drilling 4 one-inch (approximately 25 mm) holes in the old tire sidewall and cutting the DOT code out with a utility knife.

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DEALERSHIP/SUBLET TIRE SERVICE

Dealership Service Steps

1. Remove the valve stem core and deflate the affected tire(s), then remove them from the vehicle.
2. Provide replacement tire and a copy of “Sublet Service Steps” to the servicing truck tire repair facility.
3. After the wheel and tire assembly is returned from the sublet tire repair facility, reinstall it onto the vehicle. Tighten the lug nuts to 217 Nm (160 lb-ft).
4. **WARNING: BE SURE TO READ AND FOLLOW THESE INSTRUCTIONS CAREFULLY. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN SEVERE PERSONAL INJURY OR DEATH.**
   A) Prepare to inflate the tire using:
      I. Lock-on air chuck attached to a minimum of 12 feet of hose sufficient to clear the trajectory zone.
      II. Remote inflation gauge and air flow control or equivalent combination air pressure gauge and air flow control assembly. See Figure 4.
   B) Verify that all individuals are outside the trajectory area (12 feet). See Figure 5.
   C) Verify the wheel/tire assembly is bolted to the axle and the lug nuts are tightened to specification.
   D) Inflate the tire to specification (refer to Safety Certification Label located in the drivers door opening).
5. Following Step 4, adjust the inflation pressure on the remaining wheels that do not have a valve stem leak.
6. Render the old tire unusable by drilling 4 one-inch (approximately 25 mm) holes in the old tire sidewall and cutting the DOT code out with a utility knife.
7. **CAUTION: The torque on the lug nuts must be checked 160 kilometers (100 miles) AND 800 kilometers (500 miles) after tire replacement.**

   Using a suitable torque wrench, check and retighten all lug nuts to 217 Nm (160 lb-ft). See Figure 6.

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Sublet Service Steps

WARNING!

BE SURE TO READ AND FOLLOW THESE INSTRUCTIONS CAREFULLY. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN SEVERE PERSONAL INJURY OR DEATH.

SERVICING COMMERCIAL TRUCK STEEL CORD SIDEWALL TIRES

The 19.5 inch steel cord sidewall tires equipped on F-450 and F-550 Super Duty vehicles are of an ***all steel*** radial construction, and operate at inflation pressures of up to 95 psi. As such, they are NOT light truck tires and must be serviced with the same safety considerations that are used with other medium/heavy truck steel radial tires. Personnel trained, supervised, and equipped in compliance with OSHA Standard 29 CFR - 1910.177 must perform tire service.

1. Remove the old tire from the wheel.
2. Inspect the wheel for damage. Bent, cracked or otherwise damaged wheels must not be reused or repaired.
3. Replace the valve stem with either a Dill® E-600R or a Schraeder® TR-600HP valve stem.
4. **CAUTION: Do not use petroleum-based lubricants on the replacement tire. Damage to the bead area of the tire may occur.**

   Lubricate the bead of the ***new*** replacement tire provided by the dealership, using a commercially available tire lubricant. Then mount the tire on the wheel, working from the valve side.
5. **WARNING: DO NOT FULLY INFLATE THE TIRE. SEAT THE BEAD ONLY.**

   Inflate the tire with just enough air to seat the bead (20 psi maximum).
6. Install the valve stem core. DO NOT INFLATE THE TIRES ANY FURTHER.
7. Balance the wheel and tire assembly.
8. Return the wheel/tire assembly and the removed tire to the dealership.

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